

Cross-selling strategy analysis — United India Insurance

1) Business problem & objective

United India Insurance wants to increase revenue and strengthen customer relationships by **cross-selling** additional insurance products to existing customers. Cross-selling can mean:

- Selling a **new product** in addition to the current one (increasing total coverage),
- Replacing an expired product with an equivalent product (maintaining coverage), or
- Offering a smaller/newly-tailored product after a policy expires.

Primary business objectives (from the business brief):

1. **Increase revenue** from existing customers.
2. **Protect and strengthen** the customer relationship (reducing churn and increasing lifetime value).

Selling more to existing customers is cheaper than acquiring new ones, increases customer lifetime value (CLV), and raises switching costs for competitors — all important for a large insurer.

2) Dataset Overview

File name: Insurance Data.csv

Rows: 100,000

Columns: 16

This data-set contains customer-level information from United India Insurance, combining demographic, product, and behavioral data. It's designed to support cross-selling and conversion analysis — identifying which customers are most likely to purchase an additional insurance product.

Short descriptions (of columns)

Age — customer age (years).

Family_Members — number of dependents/household members.

Income — customer income (numeric).

Current_Coverage — coverage amount for the currently-held policy.

New_Coverage — coverage amount being offered/considered under a new product.

Current_Product, Current_Product_Type — what product the customer currently holds and its type.

New_Product_Type — the product/category being considered for cross-sell.

Rating — lead rating categorized as e.g., *Cold/Warm/Hot* (used for prioritization).

Converted — whether the cross-sell offer resulted in conversion (likely 0/1 or Yes/No) — this is the clearest operational target for conversion modeling.

Status — Additional flag/status field (possibly related to policy renewal or conversion)

Data Composition Summary

Total Records: 100,000 customers

Numeric Columns: 6

(Age, Family Members, Income, Current_Coverage, New_Coverage)

Categorical Columns: 10

(Gender, Marital Status, Education, Occupation, Job Title, Current Product, Current Product Type, New Product Type, Rating, Converted, Status)

Missing Value Insights

Missing values are present in a few numeric and categorical columns

```
Missing values:
Age                44
Gender             182
Marital_Status     42
Family_Members     22
Education          46
Occupation         43
Job_Title          123
Income             98
Current_Product    18
Current_Product_Type 33
Current_Coverage   49
New_Product_Type   47
New_Coverage       127
Rating             44
Converted          49
Status             0
dtype: int64
```

These are handled later in the preprocessing phase using:

Mean/Median imputation for numeric variables.

Mode imputation for categorical variables.

Group-based imputation for certain variables like **Income**, using occupation-wise averages.

Target Variables

Primary Target: 'Converted' — indicates whether a customer accepted a new insurance offer (1 = Yes, 0 = No).

Secondary Variable: 'Rating' — lead heat score (Cold/Warm/Hot), can be analyzed against 'Converted' to evaluate prediction quality or lead prioritization accuracy.

Data-set summary

Data-set provides a comprehensive foundation for cross-sell opportunity analysis, predictive modeling, and marketing segmentation. It allows both **behavioral segmentation** (who converts more easily) and **value-based targeting** (which customer segments bring higher coverage or income potential).

3)Data Preprocessing

The preprocessing phase for this project involved several systematic steps, focusing on handling missing values, correcting inconsistencies, transforming data types, and preparing both categorical and numerical variables for analysis.

4.1 Handling Missing Values

The dataset contained a few missing values across both numerical and categorical columns.

To ensure data completeness and prevent information loss, the following imputation strategies were applied:

Numerical Columns:

Columns such as **Age**, **Family Members**, **Income**, **Current_Coverage**, and **New_Coverage** were checked for missing entries.

For most numeric columns, missing values were replaced with **mean** or **median** values, depending on the distribution of each variable.

For the **Income** column specifically, missing values were imputed using **Occupation-wise averages**. This approach ensured that customers in similar professions had realistic income estimates, improving the quality of imputation.

Categorical Columns:

Categorical variables such as **Gender**, **Marital Status**, **Education**, **Occupation**, **Job Title**, **Current Product**, **Current Product Type**, **New Product Type**, and **Rating** were imputed using the **mode** (the most frequent category).

This method helped maintain category consistency and prevented artificial class imbalance.

In cases where categorical variables were interdependent (e.g., **Current Product**, **Current Product Type**, and **Current_Coverage**), the relationships between columns were carefully examined before imputation to preserve logical integrity.

Target Column — Converted:

The **Converted** column represents whether a customer accepted the new insurance offer (Yes/No).

As this column is crucial for analysis and modeling, **rows containing missing values in Converted were dropped entirely** rather than imputed.

This ensured that only valid and meaningful records were retained for downstream analysis.

4.2 Data Cleaning and Standardization

Column Name Standardization:

All column names containing spaces were replaced with underscores (**_**) for

consistency and ease of reference in code (e.g., Marital Status → Marital_Status).

Irrelevant or Redundant Columns:

Columns found to contain redundant information (such as Status, which duplicated information from Converted) were dropped after validation.

Data Type Conversion:

Categorical columns were explicitly converted to the object data type for consistent encoding.

This ensured compatibility during calculations and visualizations in later stages.

4.3 Target Variable Preparation

The **Converted** column (Yes/No) was transformed into a binary numeric format to create a modeling target variable.

Converted → 1 (Customer purchased the new product)

Not Converted → 0 (Customer did not purchase the new product)

A new column named **target** was temporarily created for this numeric representation and later standardized back to Converted for clarity.

4.4 Categorical and Numerical Feature Identification

To streamline later analysis and model preparation, two variable lists were created:

Categorical Columns (cat_cols):

['Gender', 'Marital_Status', 'Education', 'Occupation', 'Job_Title', 'Current_Product', 'Current_Product_Type', 'New_Product_Type', 'Rating']

Numerical Columns (num_cols):

['Age', 'Family_Members', 'Income', 'Current_Coverage', 'New_Coverage']

These lists were used throughout preprocessing to automate imputation, data type conversions, and future encoding processes.

4.5 Final Data Validation

After completing all preprocessing tasks, the dataset was rechecked to ensure:

- No missing values remained (`isnull().sum()` returned zero).
- Data types matched expectations (numeric vs. categorical).
- Column names were standardized and consistent.
- Each record represented a unique, valid customer.

The cleaned dataset was then saved as a new file (`Cleaned_ds.csv`) for further analysis and modeling.

Outcome:

Following preprocessing, the dataset was fully cleaned and consistent, containing 99951 unique and valid records across 15 well-defined features.

This created a robust foundation for further exploratory analysis and modeling, ensuring that results derived in the next phase would be accurate and meaningful.

4) Exploratory Analysis — Visualizations & Insights

1. Univariate Analysis

What is Univariate Analysis

Univariate analysis refers to examining **one variable at a time** to understand its distribution, range, central tendency, and variability.

It helps analysts identify patterns, detect outliers, and understand how each feature behaves independently before comparing it to other variables.

Univariate analysis allows us to:

Examine how each **numerical feature** (e.g., Age, Income, Coverage) is distributed.

Understand the **difference in shape** of these distributions between customers who **converted** and those who **did not convert**.

Detect any skewness, outliers, or natural groupings that may affect conversion probability.

Performing univariate analysis before bivariate or multivariate analysis ensures that we fully understand the **basic data structure** and **individual behavior of variables**.

Both **KDE plots** (for distribution) and **Boxplots** (for spread and outliers) were used for key numeric features against the **Conversion Status**.

Additionally, **mean values** were computed to quantify these visual observations.

Numerical Variables vs Conversion Status

The following numeric variables were analyzed:

Age, Family_Members, Income, Current_Coverage, New_Coverage

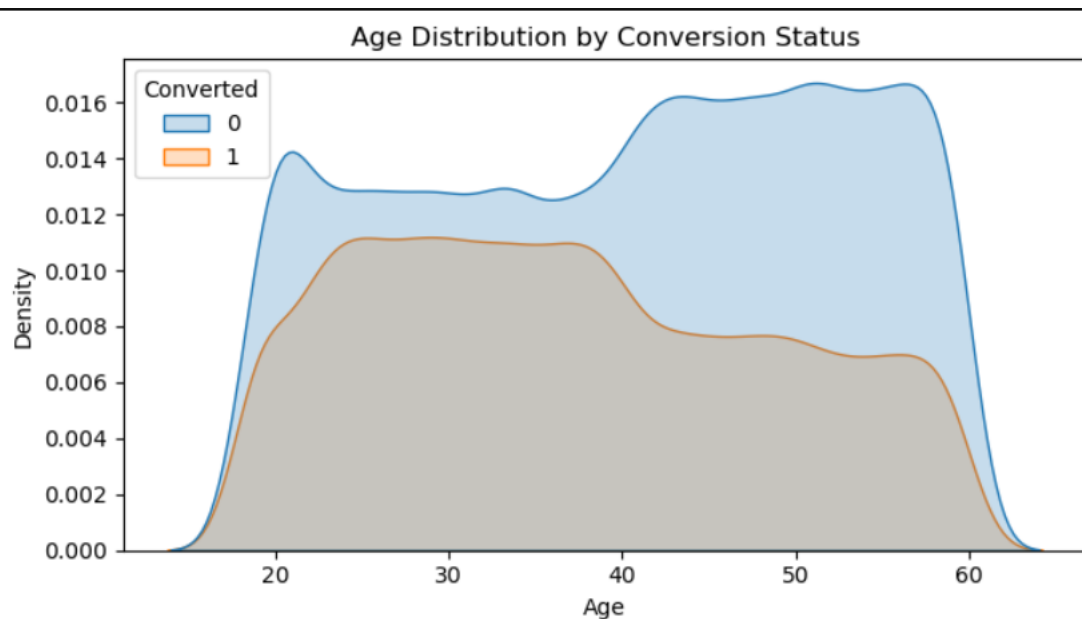
Each variable was plotted using:

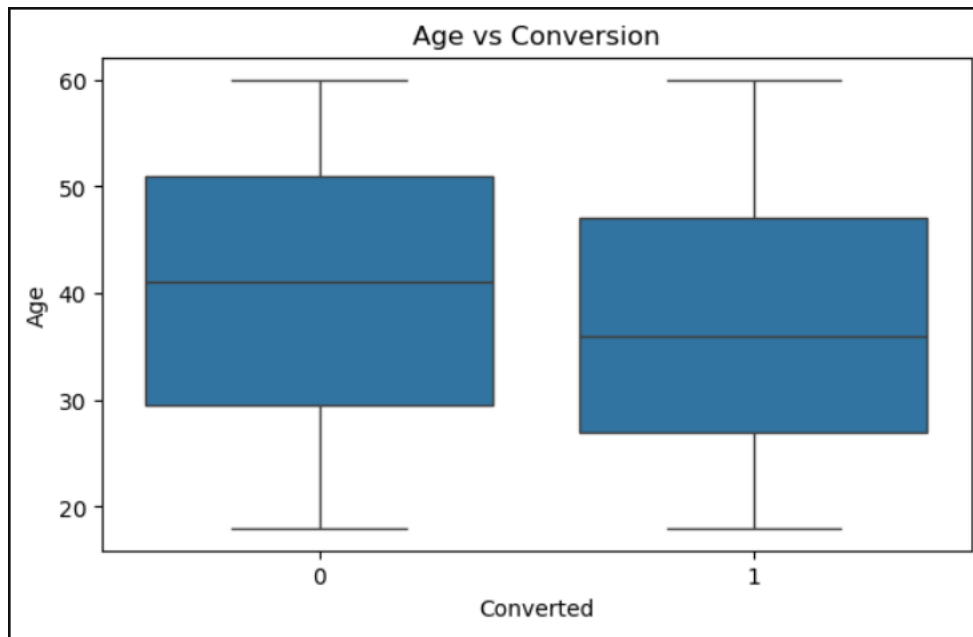
KDE Plot: To visualize distribution and shape differences between Converted and Not Converted customers.

Boxplot: To show spread, median, and outlier behavior.

Mean Values: To statistically summarize the difference between groups.

Age vs Conversion





Converted customers' age distribution peaks between **30–40 years**, while Not Converted customers are concentrated between **40–50 years**.

Boxplots show a slightly lower median and narrower spread for Converted customers, suggesting younger age groups are more responsive.

Mean Age:

Conversion Status Mean Age

Converted	36.8
Not Converted	41.9

Insight:

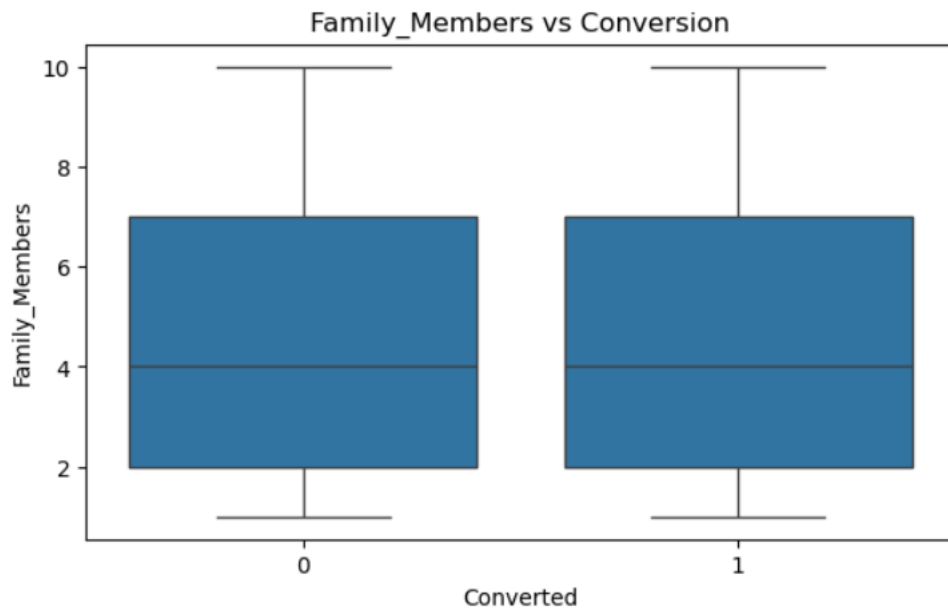
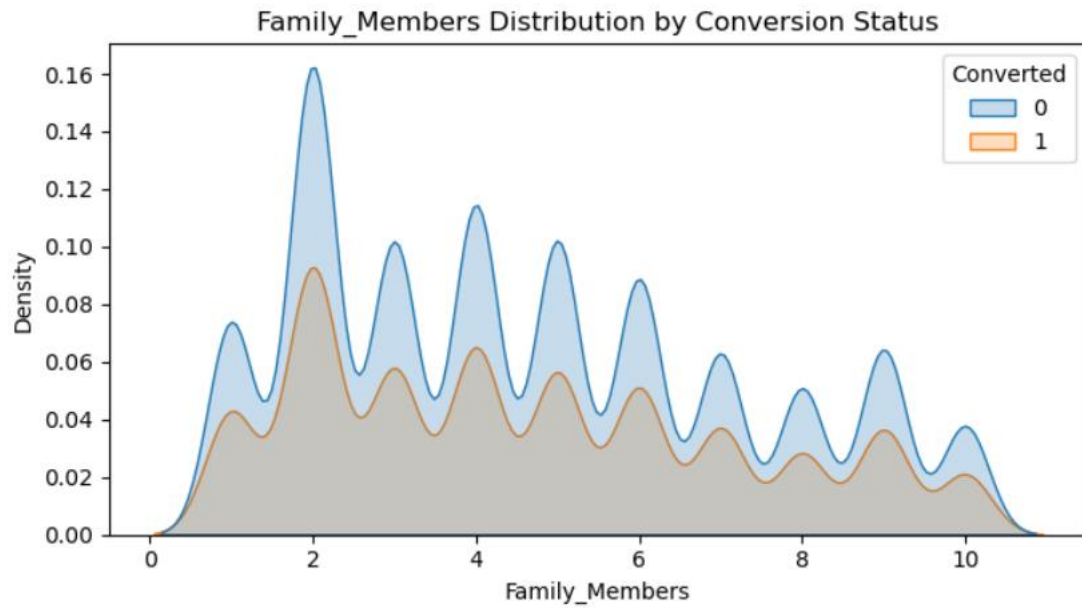
Younger customers (mostly in their 30s) tend to convert more frequently.

This may reflect increased financial awareness or first-time insurance purchases in early career stages.

Business Implication:

Focus marketing campaigns on the **30–40 age group**, particularly through digital and lifestyle-oriented channels.

Family Members vs Conversion



Both Converted and Not Converted groups show similar family size distributions, peaking at **4 family members**.

The boxplots are almost identical, with no significant outliers or shifts.

Mean Family Size:

Conversion Status Mean Family Members

Converted 4.08

Not Converted 4.02

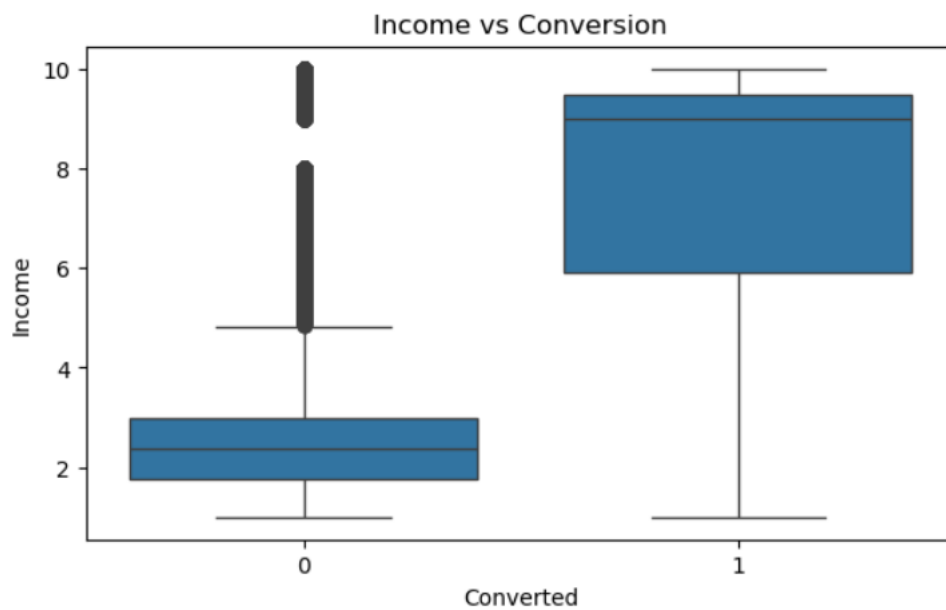
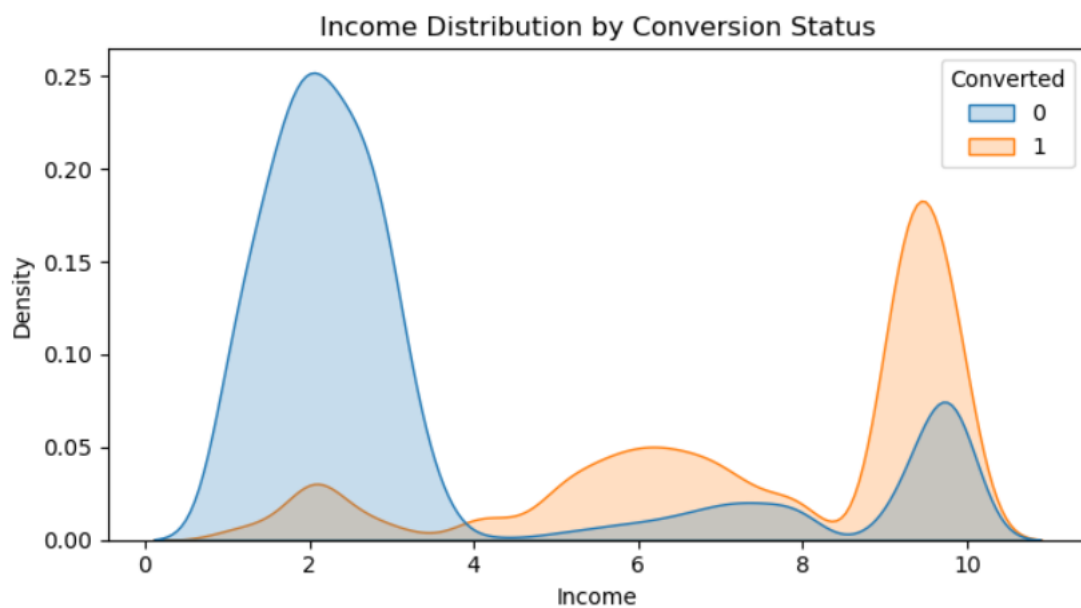
Insight:

Family size does not significantly influence conversion behavior.

Whether a person belongs to a small or large family, their likelihood of purchasing remains relatively constant.

Business Implication:

Family size can be deprioritized as a segmentation variable during campaign planning.

Income vs Conversion

The Converted group's distribution is **right-shifted**, showing that customers with **higher income levels** convert more often.

Boxplots reveal a visibly higher median for Converted leads, with a few high-income outliers.

Mean Income:

Conversion Status Mean Income

Converted ₹9,075,873

Not Converted ₹2,073,777

Insight:

Income shows a **strong positive correlation** with conversion probability.

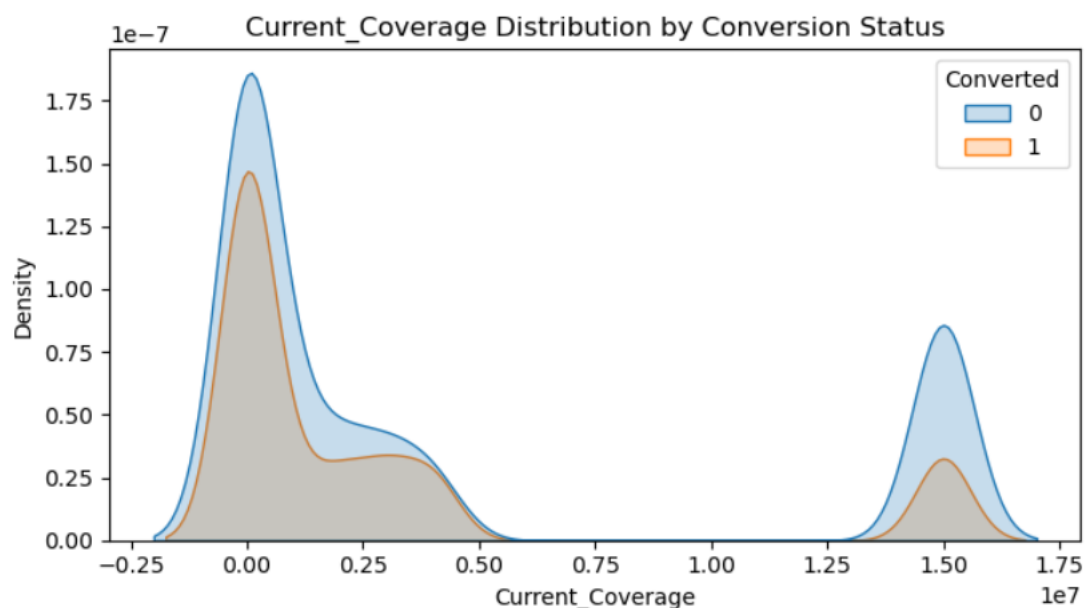
Wealthier individuals are more inclined to purchase new or additional insurance products.

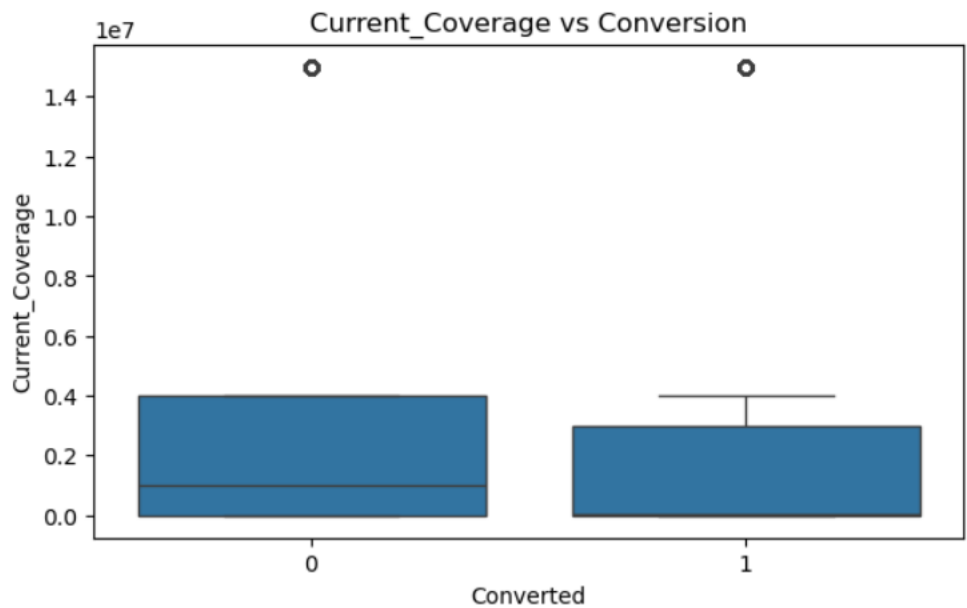
Business Implication:

High-income customers should be prioritized in targeted outreach.

Introduce **premium product bundles** or **investment-linked policies** for this segment

Current Coverage vs Conversion





Converted customers mostly cluster around **zero current coverage**, while Not Converted leads have higher existing coverage levels.

Boxplots show a much lower median and narrower spread for Converted customers.

Mean Current Coverage:

Conversion Status Mean Current Coverage

Converted ₹14,543.77

Not Converted ₹50,000.00

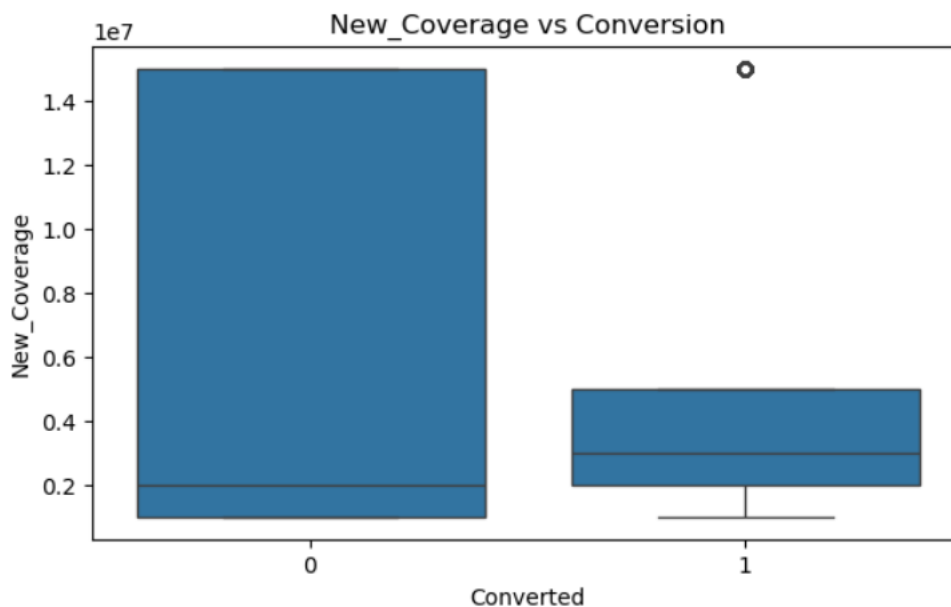
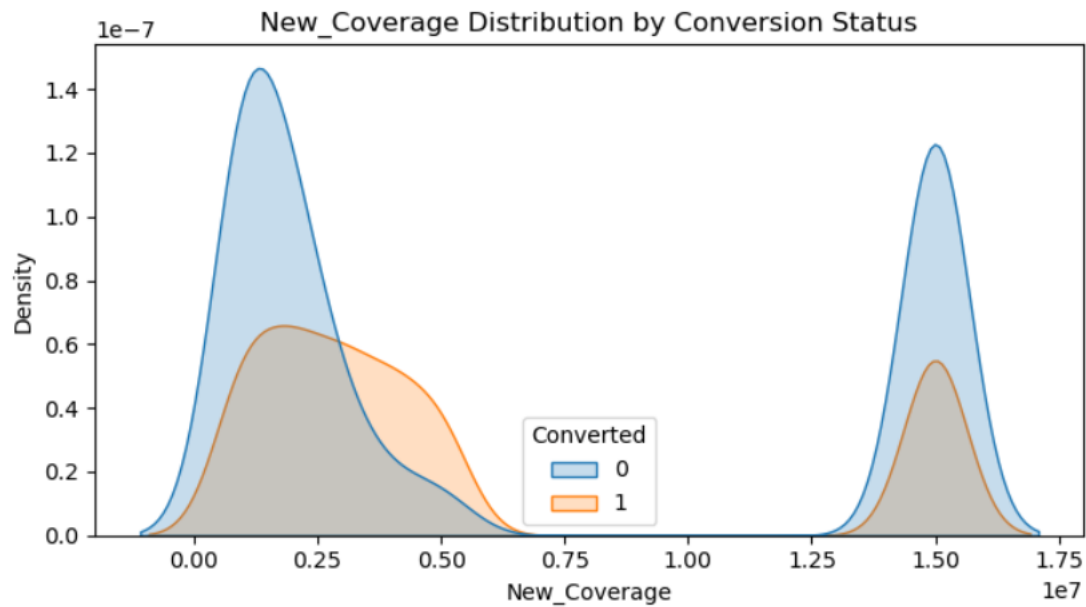
Insight:

Customers **with little or no current coverage** are more likely to accept new insurance offers, indicating a **coverage gap** that drives purchase intent.

Business Implication:

Run targeted campaigns for customers with **low or no current coverage**, positioning the product as a protection upgrade or first-time insurance solution.

New Coverage vs Conversion



Converted customers show higher density toward **larger coverage amounts**.

Boxplots confirm that Converted leads opt for **higher new coverage** on average.

Mean New Coverage:

Conversion Status Mean New Coverage

Converted ₹3, 220, 054

Not Converted ₹1, 900, 212

Insight:

Customers who convert are willing to invest in **higher-value policies**, suggesting greater confidence and perceived value.

Business Implication:

Encourage upselling and cross-selling of **high-value coverage plans** for likely converters.

This group can also be targeted for **add-on benefits** or **premium rider options**.

Categorical Variables

Objective and Approach

Categorical variables were analyzed to identify how different demographic and product-related attributes influence the **conversion status** (whether a customer accepted the new insurance product).

Two techniques were used:

Countplots — to visually compare the distribution of each category across Converted and Not Converted groups.

Mode Calculation — to identify the **most frequent (mode)** category within each group, giving a concise summary of dominant trends.

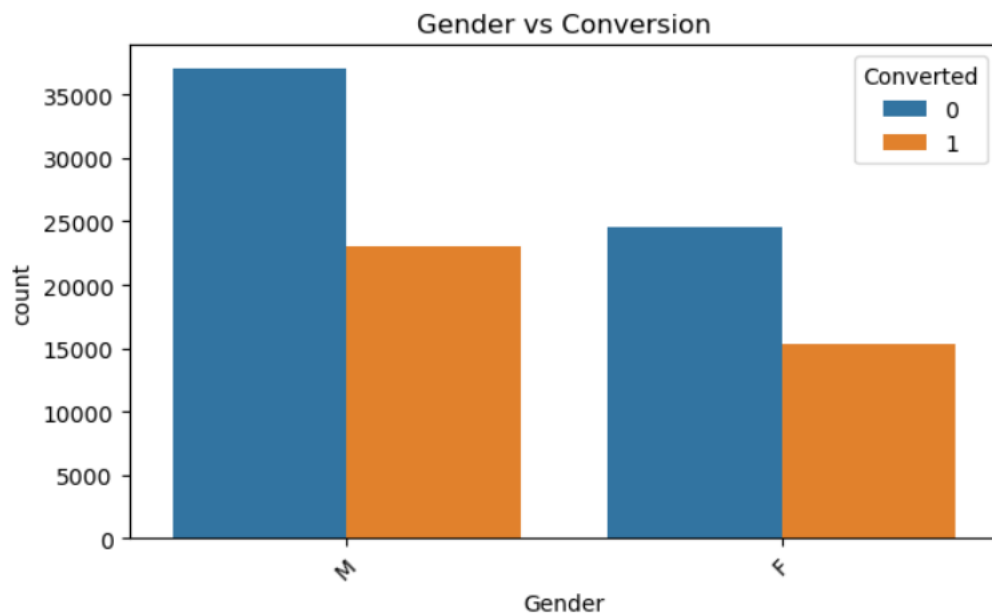
These analyses help determine which categorical characteristics are most associated with successful conversions.

Categorical Variables vs Conversion Status

The following categorical variables were examined:

Gender, Marital_Status, Education, Occupation, Job_Title, Current_Product, Current_Product_Type, New_Product_Type

Gender vs Conversion



Visualization: Countplot comparing gender distribution across Converted and Not Converted groups.

Observation:

Both genders have nearly equal participation rates in conversions.

The **mode** for both groups is 'Male'.

Converted: Male → 60%, Female → 40%

Not Converted: Male → 61%, Female → 39%

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Mode Summary:

Conversion Status Most Frequent Gender

Converted Male

Not Converted Male

Insight:

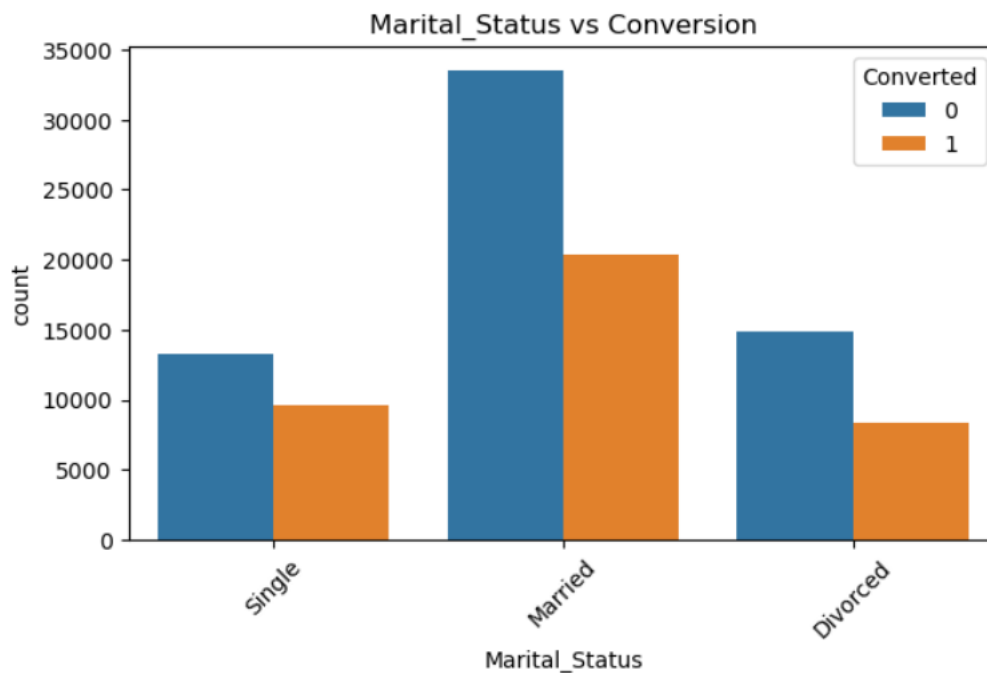
Gender does not have a significant impact on conversion.

The proportions of Male and Female customers converting are almost identical.

Business Implication:

Marketing strategies can remain gender-neutral — focus should instead be on other demographic or behavioral attributes.

Marital Status vs Conversion



Observation:

Single customers show higher conversion counts than Married customers.

Mode for both Converted and Not Converted is '**Single**', but the proportion of singles among Converted customers is noticeably higher.

Mode Summary:

Conversion Status Most Frequent Marital Status

Converted Single

Not Converted Single

Insight:

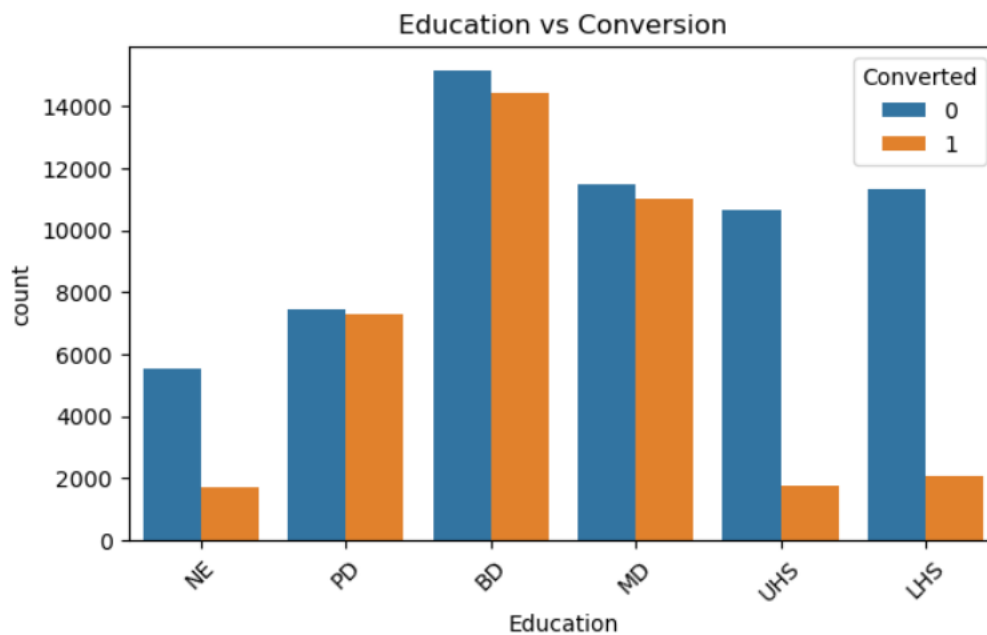
Single customers demonstrate higher responsiveness toward new insurance offers. They may be more financially independent and open to new investment or protection plans.

Business Implication:

Target single professionals with personalized savings or protection plans.

Married customers may respond better to family- or child-benefit-oriented packages.

Education vs Conversion



Observation:

Majority of Converted customers belong to higher education categories (MD, PD, or BD).

The mode for Converted customers is '**MD**', while for Not Converted it is '**BD**'.

Higher educational qualifications show slightly higher conversion ratios.

Mode Summary:

Conversion Status Most Frequent Education

Converted MD

Not Converted BD

Insight:

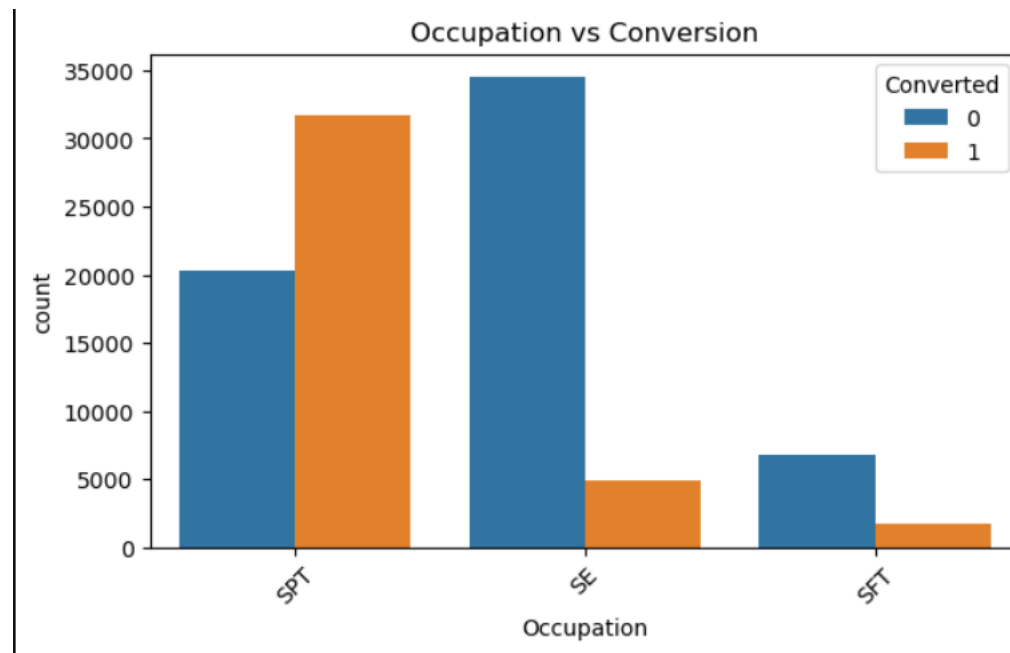
Higher education levels correlate positively with conversion.

Educated customers likely have a stronger understanding of the importance of insurance and investment.

Business Implication:

Promotional content emphasizing product returns and benefits should be directed at well-educated segments.

Occupation vs Conversion



Observation:

Certain occupation categories like ‘SPT’ (**Support Professionals**) dominate among Converted leads.

The mode for Converted group is ‘SPT’, while for Not Converted it is ‘SFT’ (**Service Field Technicians**).

Converted customers are more often from professional or technical fields.

Mode Summary:

Conversion Status Most Frequent Occupation

Converted SPT

Not Converted SFT

Insight:

Occupation is a strong differentiator.

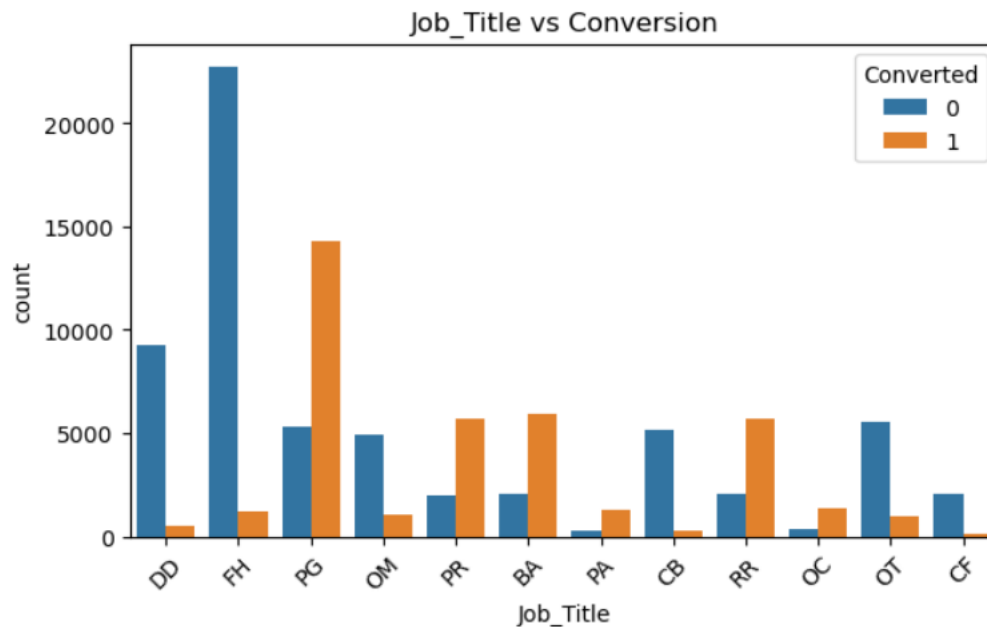
Professional and support roles are more likely to purchase new insurance products.

Business Implication:

Focus cross-sell campaigns on professionals (SPT) with stable income levels.

Explore alternate product types for field-level or service-based occupations where conversions are lower.

Job Title vs Conversion



Observation:

Job titles such as **‘PA’ (Personal Assistant)**, **‘OC’ (Officer Category)**, and **‘PR’ (Public Relations)** appear frequently among Converted customers.

The mode for Converted group is **‘PA’**, while for Not Converted it is **‘SFT’** or other mid-level categories.

Mode Summary:

Conversion Status Most Frequent Job Title

Converted PA

Not Converted SFT

Insight:

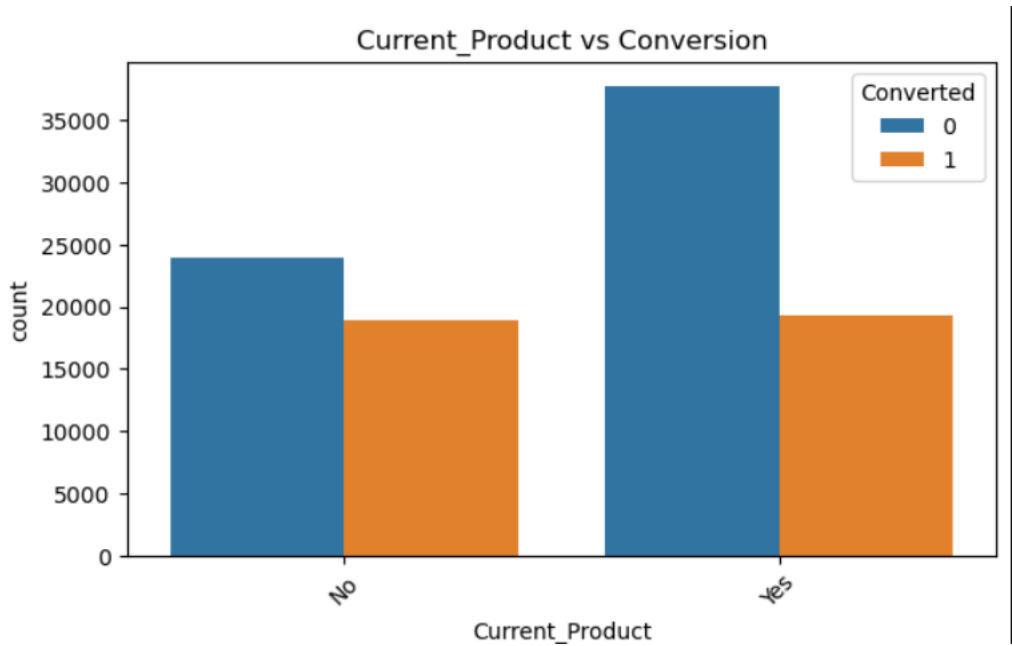
Certain job titles within organizations correlate with higher conversions — possibly due to stable salaries or awareness of financial planning.

Business Implication:

Job title can be a micro-segmentation variable in lead scoring.

Target middle- to senior-level professionals with tailored insurance packages.

Current Product vs Conversion



Observation:

Customers who already hold a **‘Health’ or ‘Life’** product show moderate conversion, but those with **no current product** convert more.

The mode for Converted customers is **‘None’**, while Not Converted is **‘Health’**.

Mode Summary:

Conversion Status	Most Frequent Current Product
Converted	None
Not Converted	Health

Insight:

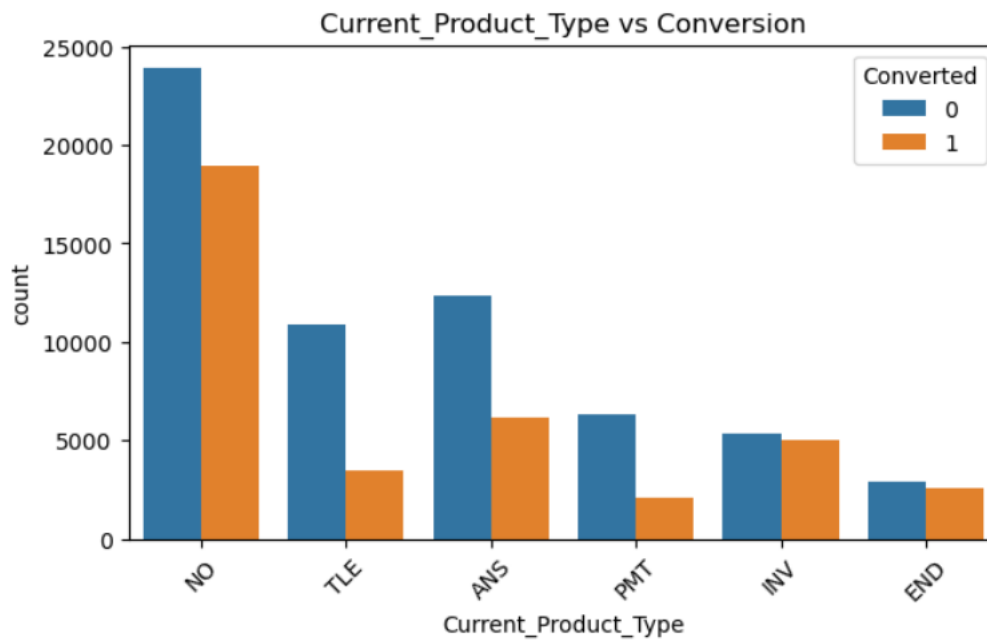
Customers with no existing product are more receptive to purchasing new ones. Those already holding products may feel adequately covered or have reduced need.

Business Implication:

Identify customers without any active policy for first-time acquisition campaigns.

For existing policyholders, use cross-sell strategies or loyalty upgrades.

Current Product Type vs Conversion



Observation:

Converted customers most frequently belong to **‘END’ (Endowment)** or **‘INV’ (Investment)** product types.

The mode for Converted is **‘END’**, while for Not Converted it is **‘INV’**.

Mode Summary:

Conversion Status Most Frequent Current Product Type

Converted END

Not Converted INV

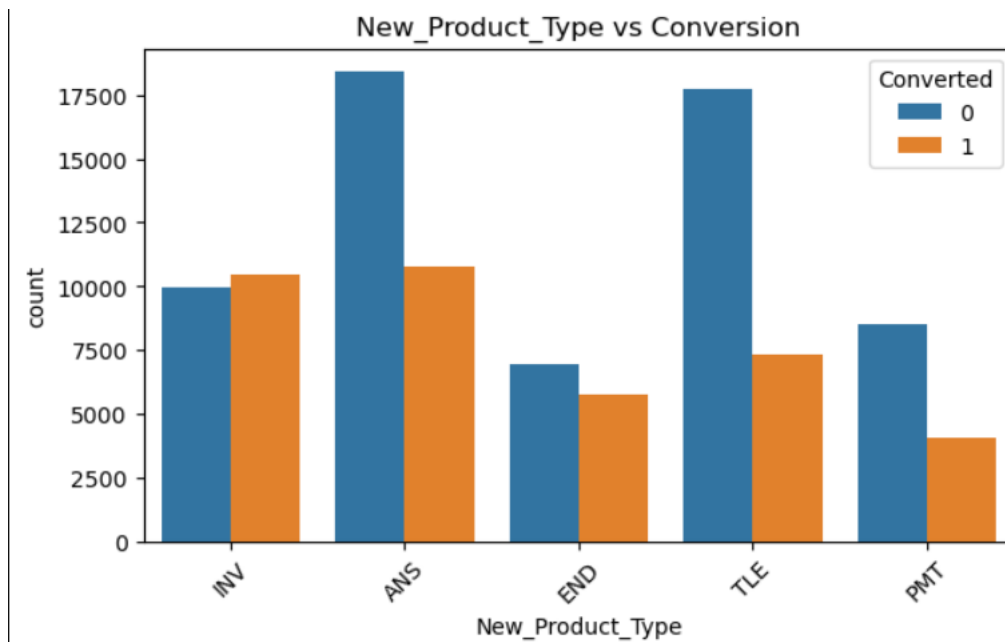
Insight:

Customers who already own Endowment plans are more likely to convert again, possibly because they value savings-oriented insurance.

Business Implication:

Endowment policyholders can be targeted with top-up or maturity benefit enhancement plans.

New Product Type vs Conversion



Observation:

Converted customers show higher counts in ‘INV’ (Investment) and ‘END’ (Endowment) categories.

The mode for Converted is ‘INV’, while Not Converted is ‘ANS’ (Annuity).

Mode Summary:

Conversion Status Most Frequent New Product Type

Converted INV

Not Converted ANS

Insight:

Investment and Endowment products have the highest success rates, while Annuity-type plans see lower interest.

Business Implication:

Prioritize marketing and upselling of Investment and Endowment-type products, as they align with customer preferences and yield higher conversions.

Numerical Variables by Rating

Objective and Approach

After comparing numerical variables with the conversion outcome, the next step involves examining how these variables differ across the **lead ratings** — *Cold*, *Warm*, and *Hot*.

This helps determine whether the **lead scoring system** (Rating) is consistent with key demographic and financial features.

Techniques used:

KDE Plots: To observe how distributions shift across rating levels.

Boxplots: To identify median and spread differences.

Mean Calculation: To quantify these trends numerically.

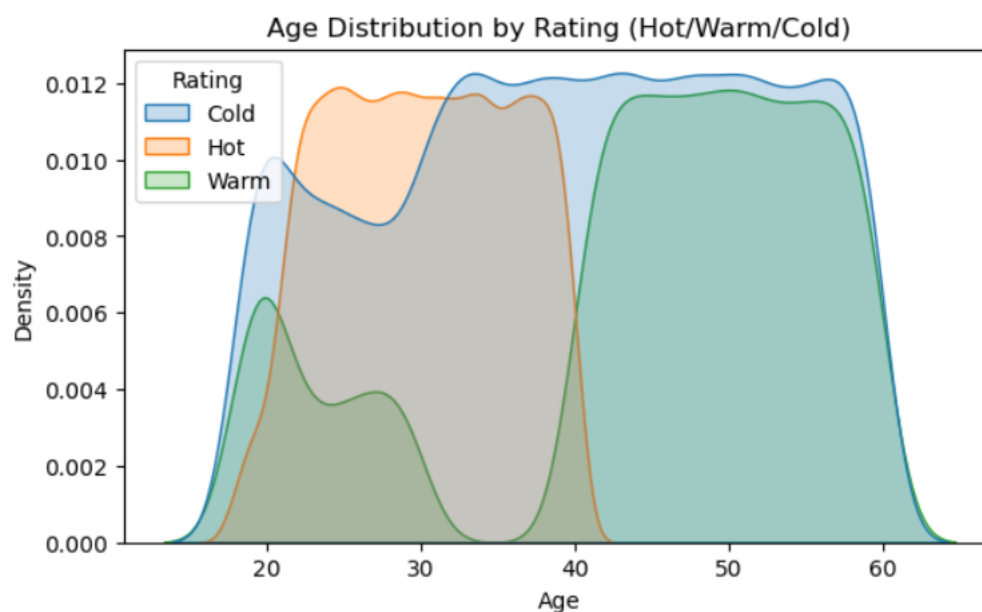
9.2 Numerical Variables vs Rating

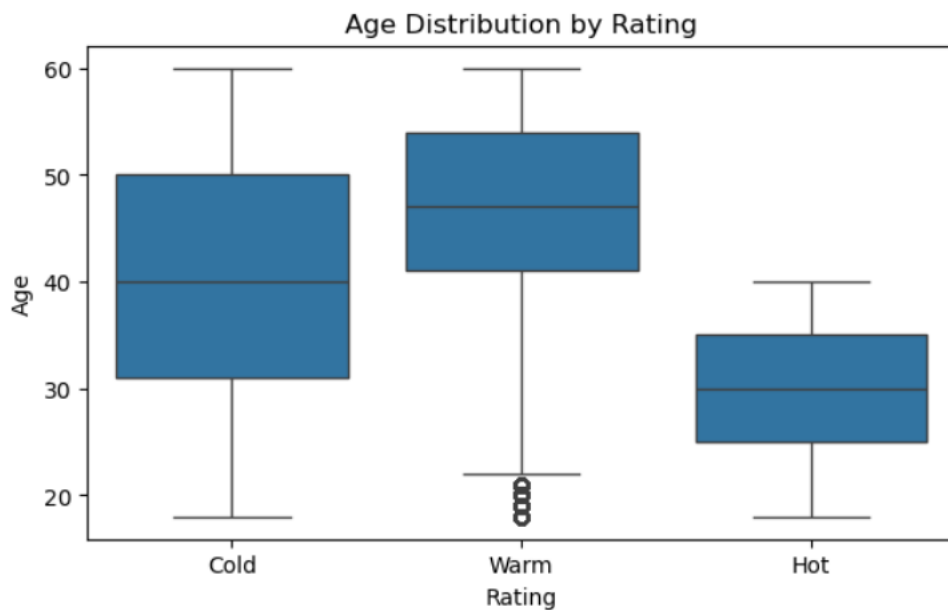
The following numeric features were analyzed:

Age, Family_Members, Income, Current_Coverage, New_Coverage

Age vs Rating

Visualization: KDE and Boxplot of Age grouped by Rating.





Observation:

The **Hot** leads' age distribution peaks around **30–35 years**, while **Cold** leads cluster around **40–45 years**.

The **Warm** category lies in between, showing a gradual age transition from Cold to Hot.

Boxplots confirm a decreasing median from Cold → Warm → Hot.

Mean Age:

Rating Mean Age

Cold 40.2

Warm 36.7

Hot 30.9

Insight:

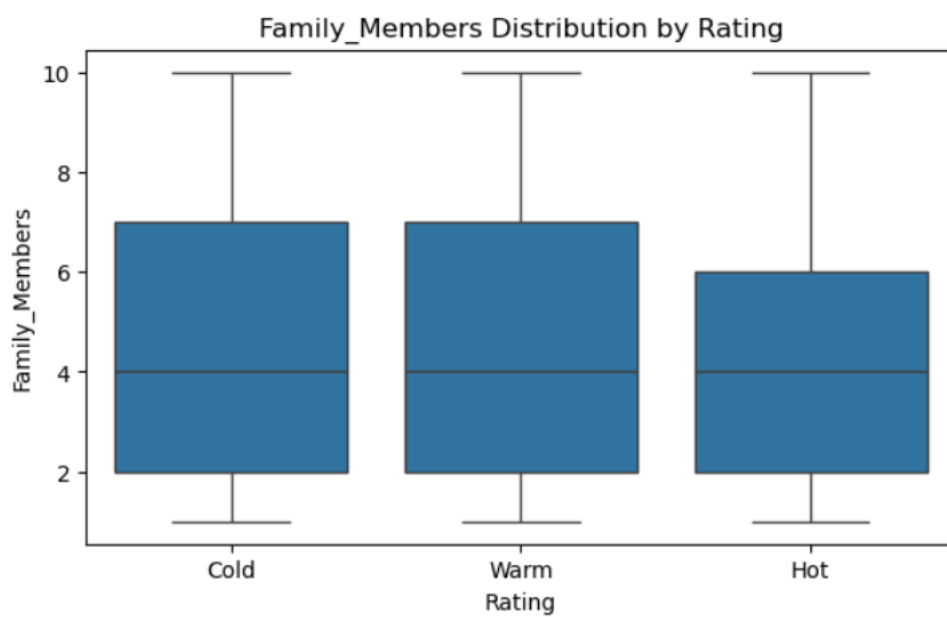
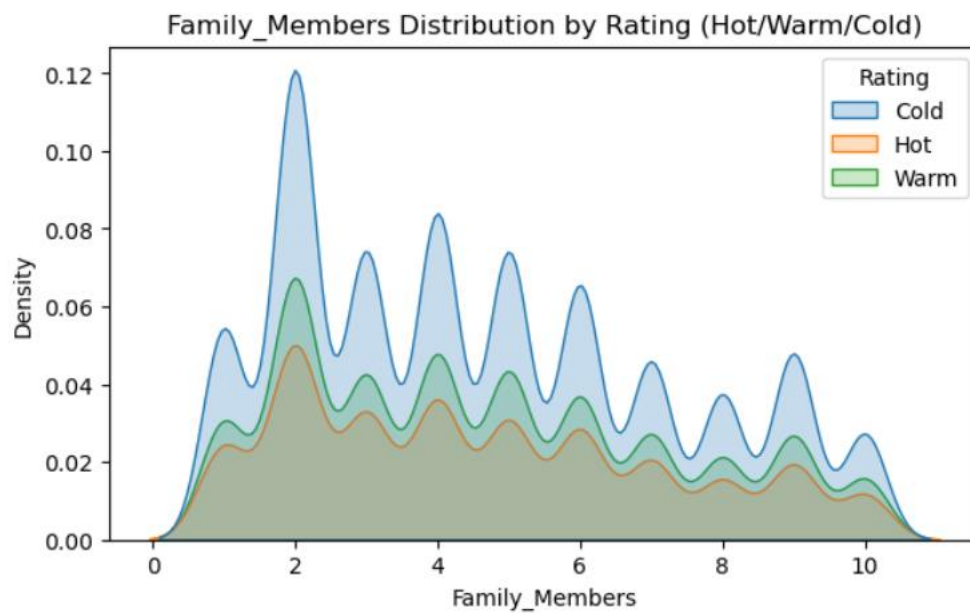
Younger individuals are more likely to be rated as Hot leads.

The rating model effectively captures **age as a signal of higher conversion potential**.

Business Implication:

Marketing strategies should emphasize **digital-first campaigns** and online communication channels targeted at the **30–35 age group** — the most promising lead segment.

Family Members vs Rating



Observation:

Family size distributions are similar across all three rating categories.

Mean values are almost constant with minimal deviation.

Mean Family Members:

Rating Mean Family Members

Cold 4.03

Rating Mean Family Members

Warm 4.06

Hot 4.01

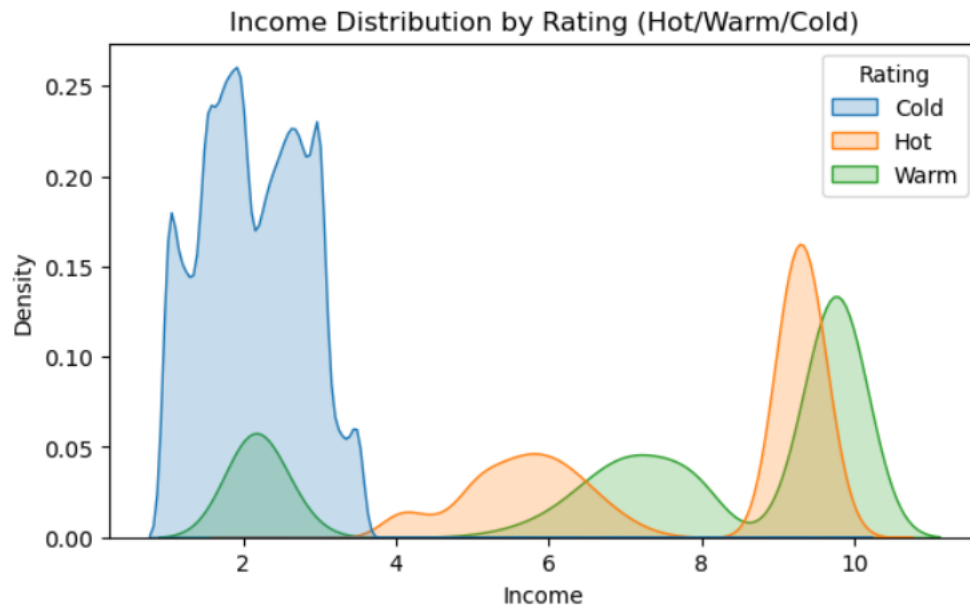
Insight:

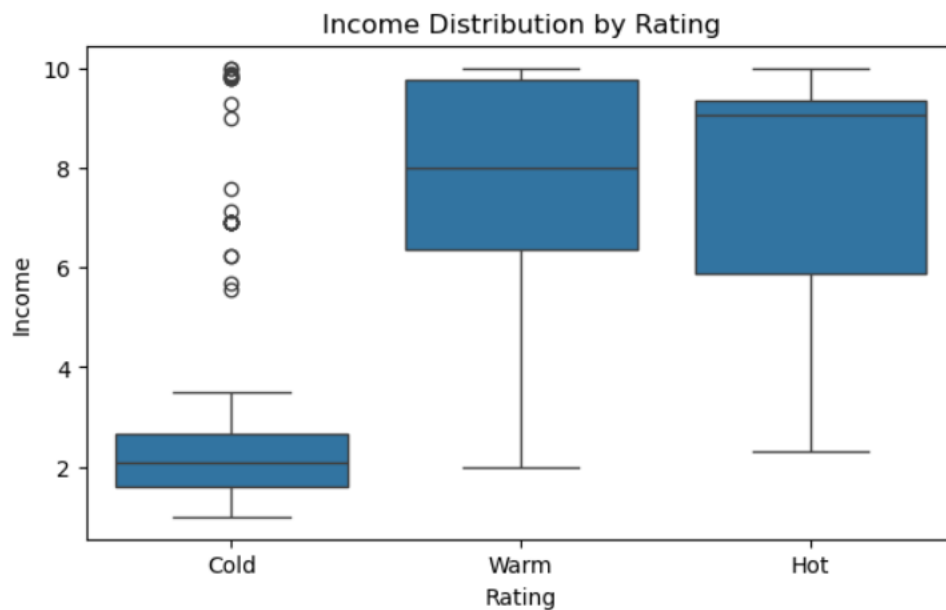
Family size does not influence lead rating or conversion probability.
All lead categories have similar household sizes.

Business Implication:

Exclude “family size” from lead scoring and segmentation models — it adds little predictive value.

Income vs Rating





Observation:

The KDE plot shows a **strong rightward shift** for Hot leads, indicating higher income levels.

Boxplots show visibly higher median and wider spread for Hot leads.

Income increases consistently from Cold → Warm → Hot.

Mean Income:

Rating Mean Income (₹)

Cold 2,073,777

Warm 8,000,000

Hot 9,075,873

Insight:

Income is a major driver in the rating model.

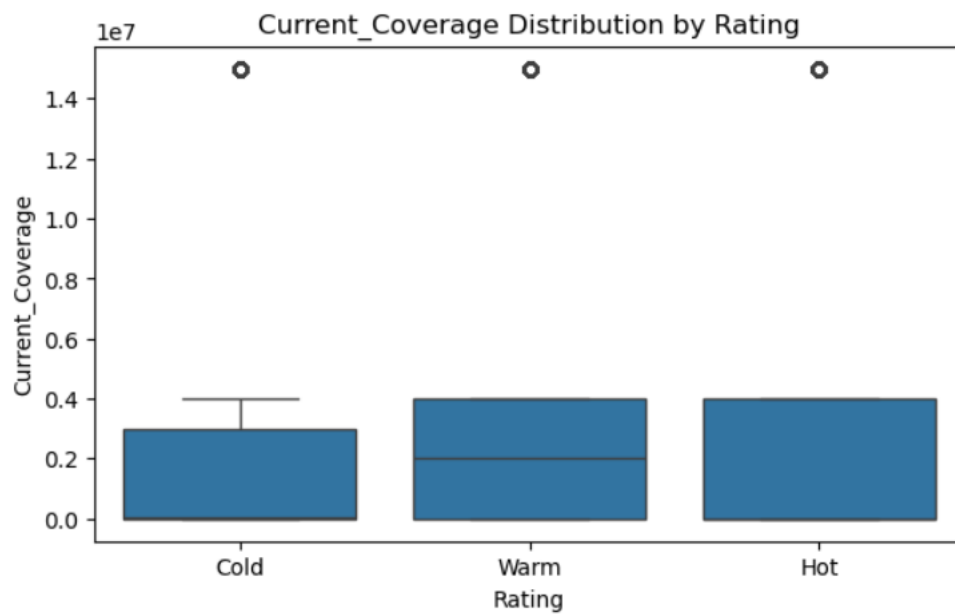
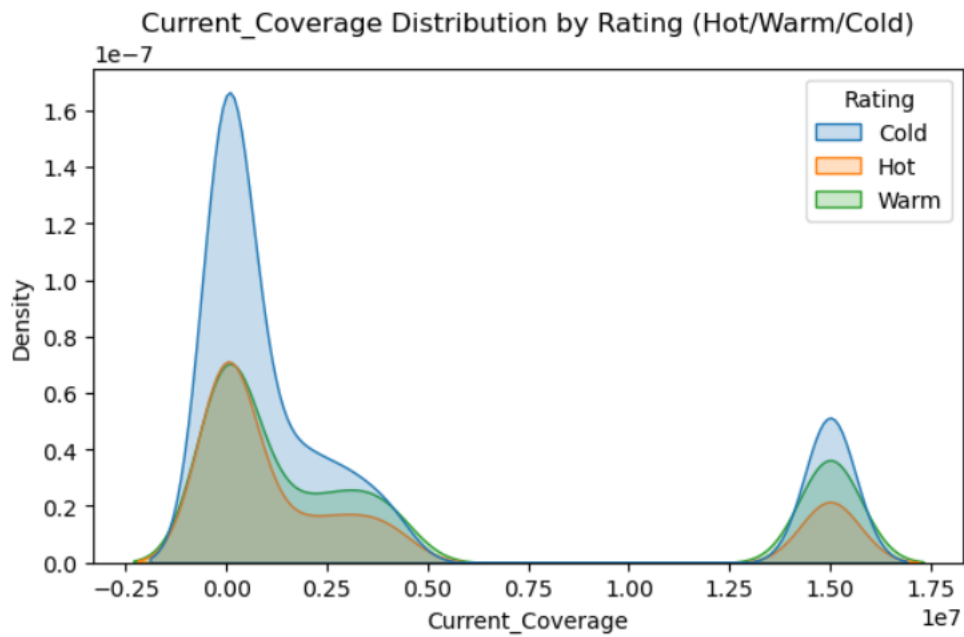
High-income customers are predominantly rated as **Hot** leads, reflecting their greater purchasing power.

Business Implication:

Hot leads represent **high-value customers**.

Offer them premium or investment-linked insurance plans to maximize revenue potential.

Current Coverage vs Rating



Observation:

Cold leads tend to have **higher existing coverage**, while Hot leads have **minimal or no current coverage**.

The density of zero coverage is highest for Hot leads.

Mean values drop significantly from Cold → Warm → Hot.

Mean Current Coverage:

Rating Mean Current Coverage (₹)

Cold 50,000

Warm 20,000

Hot 14,500

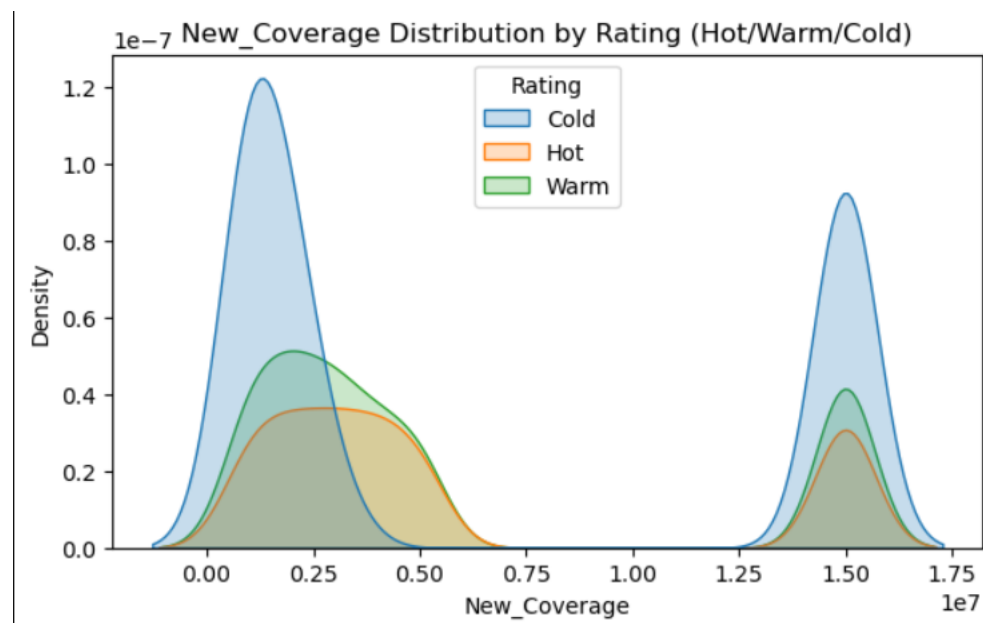
Insight:

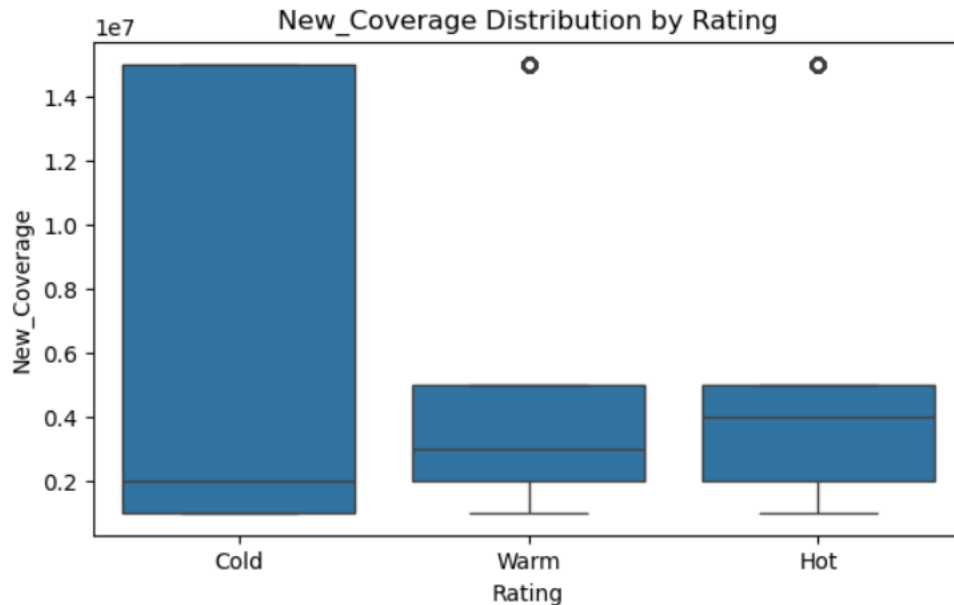
Leads with little or no current coverage are rated as Hot — suggesting the model correctly prioritizes **underinsured customers** who are more open to new products.

Business Implication:

Hot leads should be targeted with **first-time insurance acquisition** or **coverage enhancement** campaigns.

New Coverage vs Rating





Observation:

The distribution for Hot leads peaks at higher coverage values, while Cold leads are concentrated in lower coverage ranges.

The boxplots show a rising median across Cold → Warm → Hot categories.

Mean New Coverage:

Rating Mean New Coverage (₹)

Cold 2,000,000

Warm 3,000,000

Hot 4,000,000

Insight:

Hot leads are being offered or choosing **larger coverage values**, reflecting higher trust, financial capacity, and interest in more comprehensive products.

Business Implication:

For Hot leads, focus on **high-value or investment-linked products**, while offering **lower coverage or affordable plans** for Warm and Cold leads.

Categorical Variables by Rating

Objective and Approach

This analysis explores how **categorical attributes** such as demographics, occupation, and product types relate to the **lead rating** (Cold, Warm, Hot).

The goal is to understand whether specific customer categories are more likely to be high-potential leads.

Techniques used:

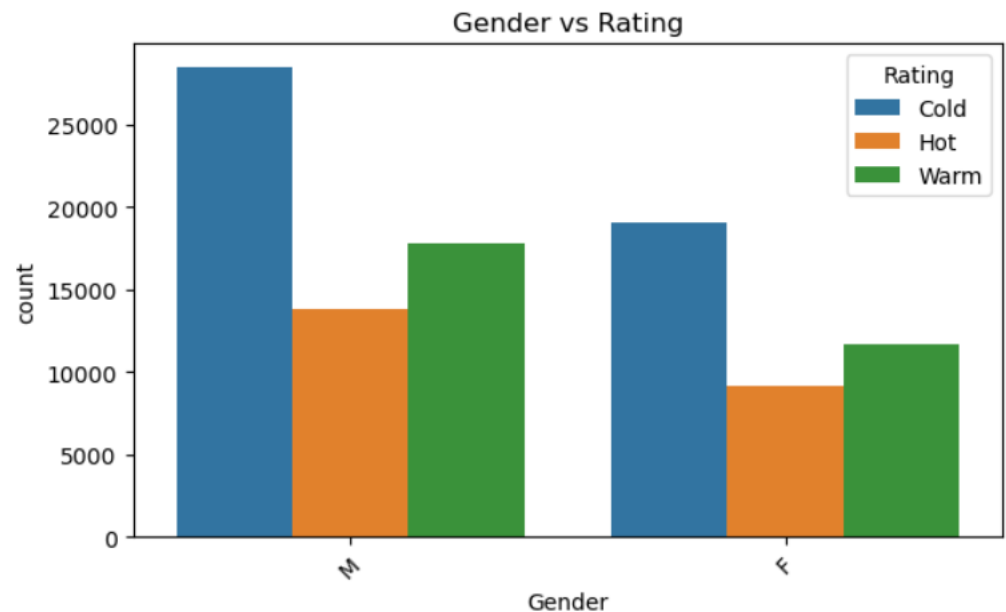
Countplots: Displayed the distribution of categories across rating levels to visually assess trends.

Mode Summary: Computed the most frequent (mode) category for each rating group to highlight dominant patterns.

Categorical Variables vs Rating

The following categorical columns were analyzed:
Gender, Marital_Status, Education, Occupation, Job_Title, Current_Product, Current_Product_Type, and New_Product_Type.

Gender vs Rating



Observation:

The distribution of males and females remains nearly uniform across Cold, Warm, and Hot ratings.

Mode: ‘Male’ for all three rating groups.

Mode Summary:

Rating Most Frequent Gender

Cold Male

Warm Male

Hot Male

Insight:

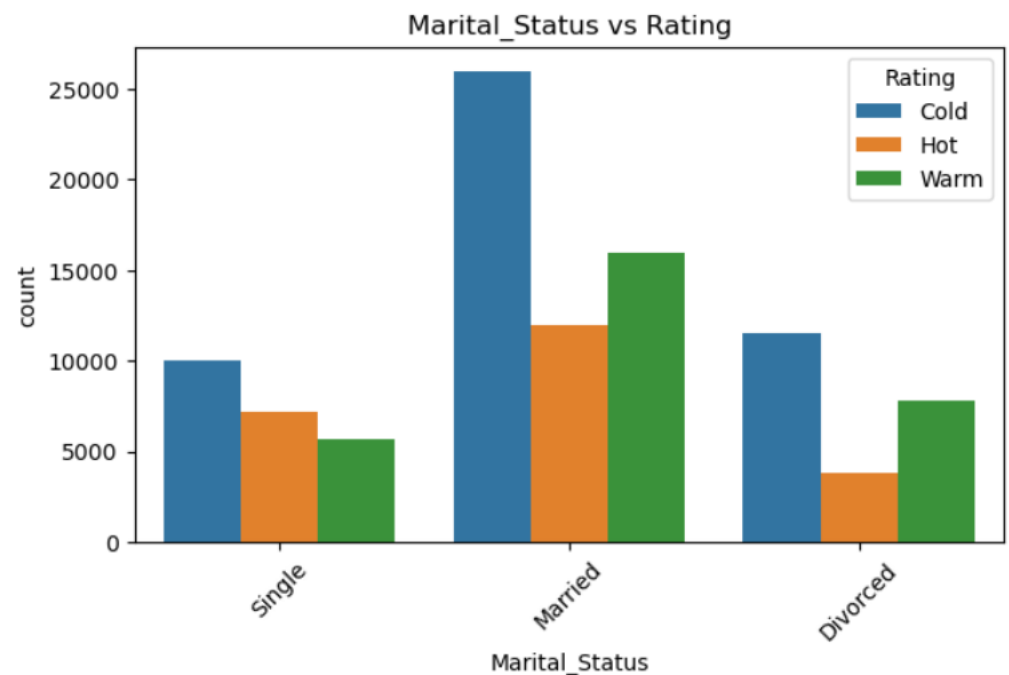
Gender has **minimal influence** on lead scoring.

The rating system does not exhibit bias or differentiation by gender.

Business Implication:

Gender-neutral marketing strategies are appropriate; focus efforts on variables with stronger predictive power.

Marital Status vs Rating



Observation:

Single customers dominate the Hot and Warm rating categories.

Married individuals are more concentrated within Cold leads.

Mode: ‘Single’ across all rating groups.

Mode Summary:

Rating Most Frequent Marital Status

Cold Single

Warm Single

Hot Single

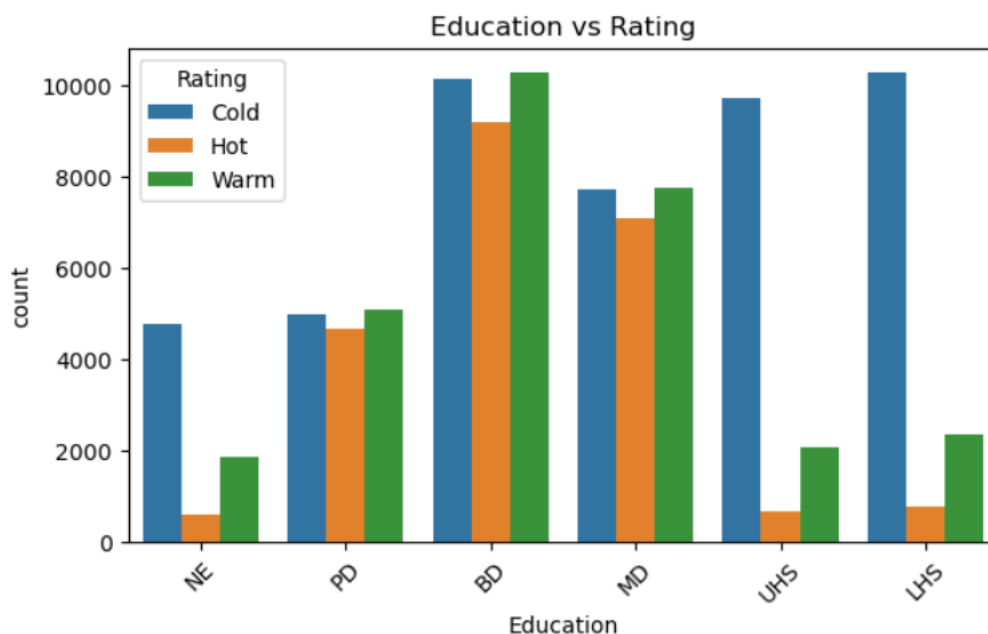
Insight:

Single customers tend to score higher on the rating scale — they are more likely to be financially independent and open to new product offers.

Business Implication:

Design marketing communication emphasizing independence, investment growth, and early financial planning for single individuals.

Education vs Rating



Observation:

Education levels such as ‘**MD**’ (Master’s Degree) and ‘**PD**’ (Post-Doctorate/Professional Degree) are more prevalent among Hot leads.

Lower education levels dominate in Cold leads.

Mode Summary:

Rating Most Frequent Education

Cold BD

Warm MD

Hot MD

Insight:

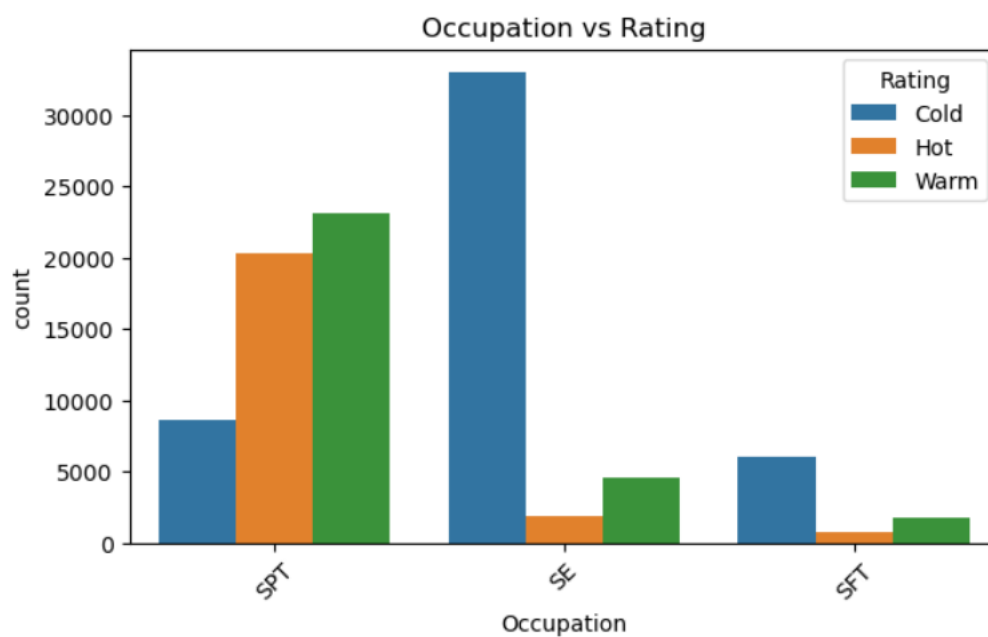
Higher education correlates positively with better lead ratings.

Educated customers are more aware of insurance benefits and exhibit stronger financial decision-making.

Business Implication:

For Hot and Warm segments, emphasize **return on investment, policy flexibility, and tax benefits** in campaign messaging.

Occupation vs Rating



Observation:

‘SPT’ (Support Professionals) and ‘SE’ (Self-Employed) categories dominate Hot leads.

‘SFT’ (Service Field Technicians) are common among Cold leads.

Mode Summary:

Rating Most Frequent Occupation

Cold SFT

Warm SE

Hot SPT

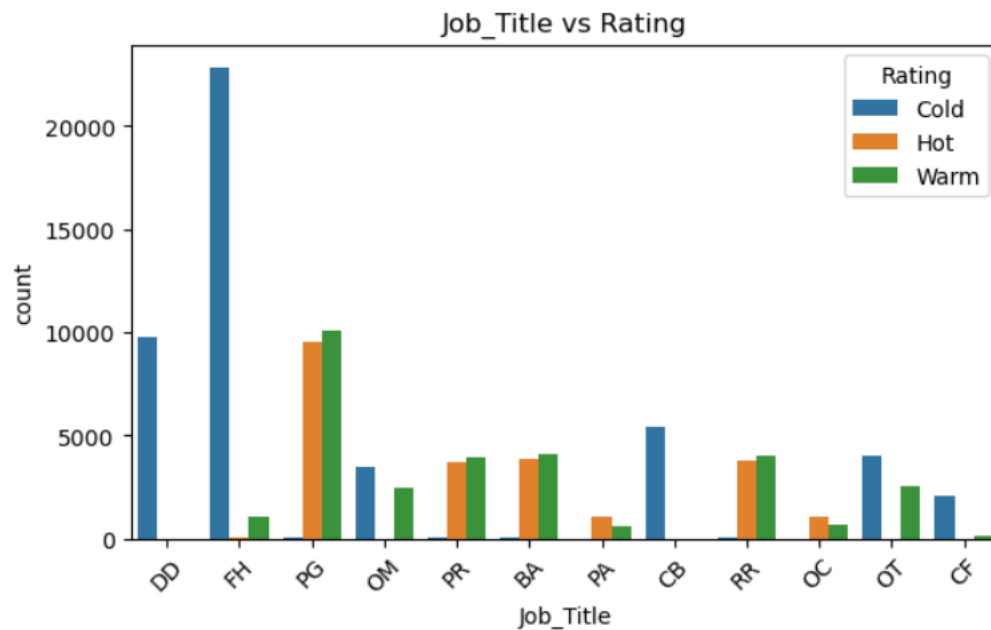
Insight:

Occupations tied to **professional or self-employed roles** tend to receive higher lead ratings, reflecting stronger purchasing capacity.

Business Implication:

Prioritize **SPT** and **SE** profiles in sales outreach, and refine rating criteria for lower-performing occupational groups.

Job Title vs Rating



Observation:

‘PA’ (Personal Assistant) and ‘OC’ (Officer Category) appear frequently among Hot leads.

‘SFT’ remains dominant in Cold leads.

Mode Summary:

Rating Most Frequent Job Title

Cold SFT

Rating Most Frequent Job Title

Warm OC

Hot PA

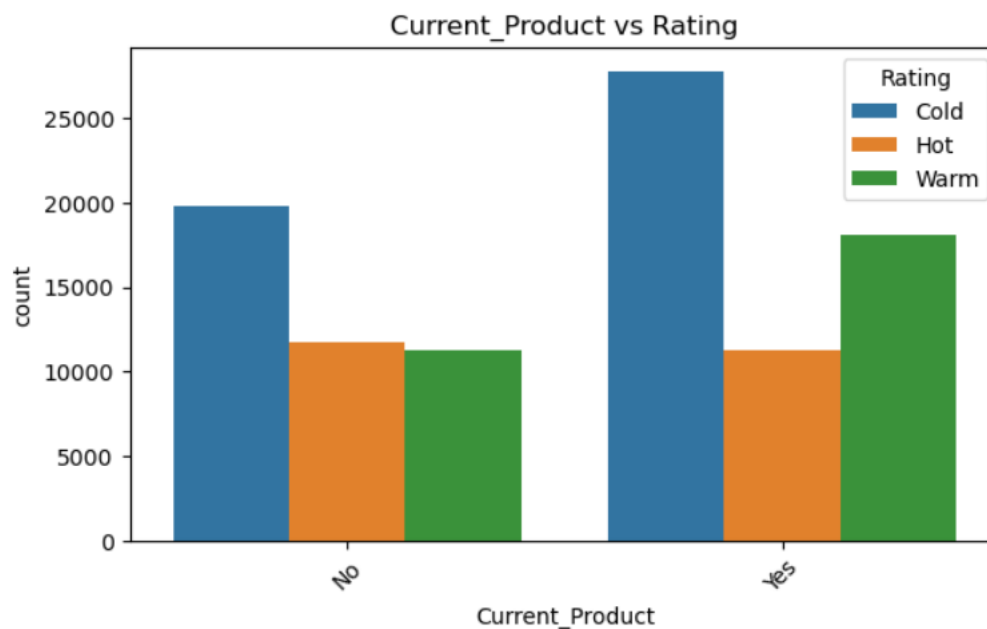
Insight:

Job title level directly correlates with lead quality. Mid- to senior-level professionals (e.g., Officers, Assistants) are more likely to be rated Hot.

Business Implication:

Use job title hierarchy as a segmentation cue in lead prioritization and personalized campaign messaging.

Current Product vs Rating



Observation:

Customers with **no existing product** dominate Hot leads, while those holding **Health or Life products** are more common in Cold leads.

Mode Summary:

Rating Most Frequent Current Product

Cold Health

Warm Life

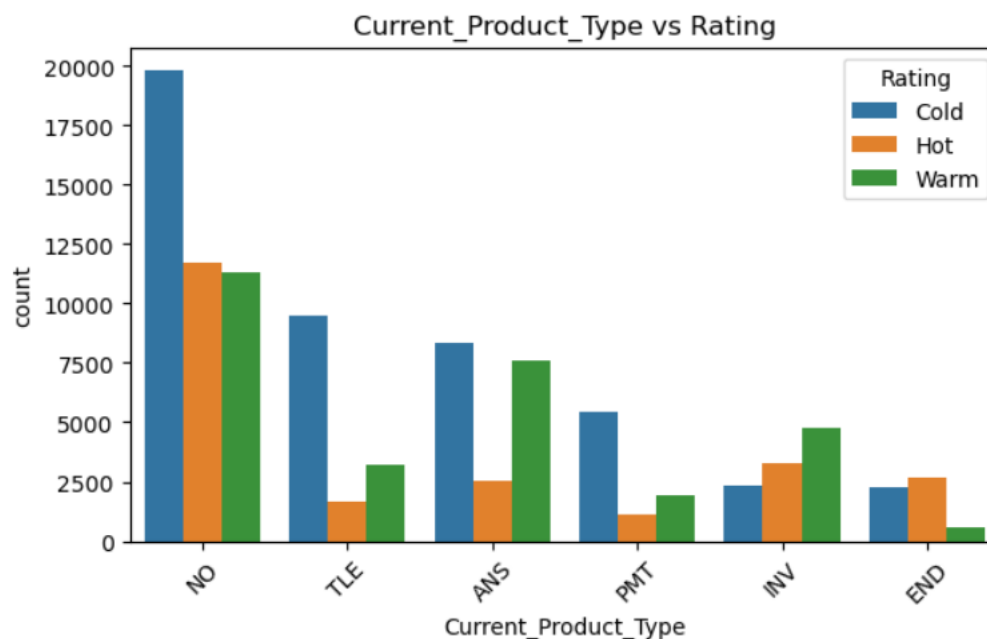
Hot None

Insight:

The rating system successfully identifies **underinsured customers** (no current policy) as high-potential (Hot) leads.

Business Implication:

Deploy **first-time buyer campaigns** for customers without existing coverage and **cross-sell programs** for those with current Health or Life policies.

Current Product Type vs Rating**Observation:**

Product types 'INV' (Investment) and 'END' (Endowment) dominate Warm and Hot leads, whereas 'ANS' (Annuity) appears more frequently in Cold leads.

Mode Summary:

Rating Most Frequent Current Product Type

Cold ANS

Warm INV

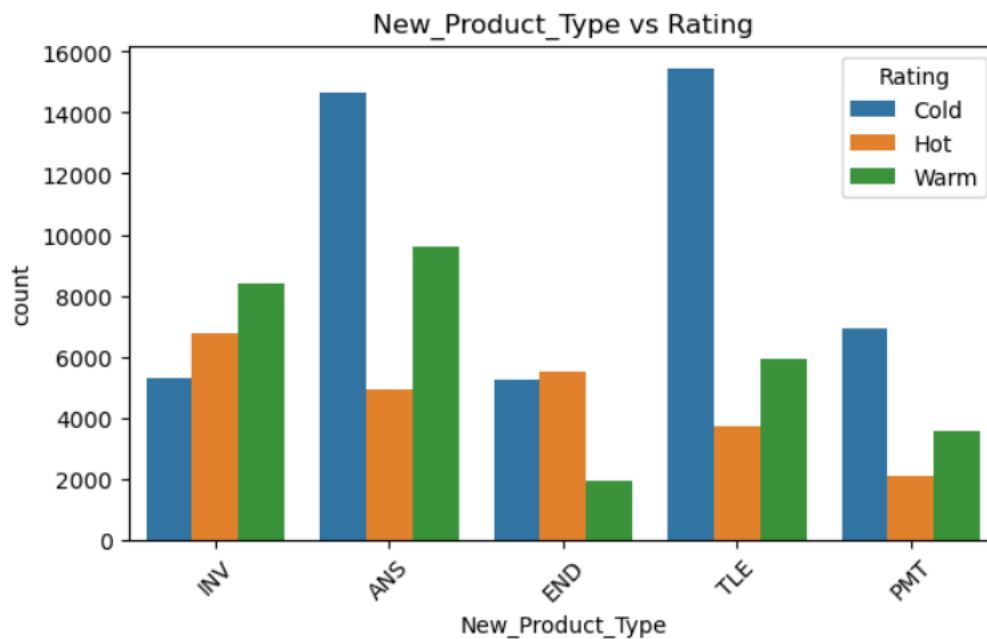
Hot END

Insight:

The shift from ANS → INV → END across Cold → Warm → Hot indicates that **savings- and investment-oriented policies** align with stronger lead scores.

Business Implication:

Prioritize customers owning Endowment or Investment-based products for premium up-sell campaigns.

New Product Type vs Rating**Observation:**

‘INV’ (Investment) dominates the Hot category, followed by ‘END’ (Endowment) in Warm leads and ‘ANS’ (Annuity) in Cold leads.

Mode Summary:

Rating Most Frequent New Product Type

Cold ANS

Warm END

Hot INV

Insight:

Investment-oriented products are consistently associated with the highest lead ratings, reinforcing their attractiveness among high-potential customers.

Business Implication:

Focus sales resources on marketing **Investment (INV)** products to Hot leads.

Comparative Analysis — Rating, Conversion, Product, and Occupation

Objective

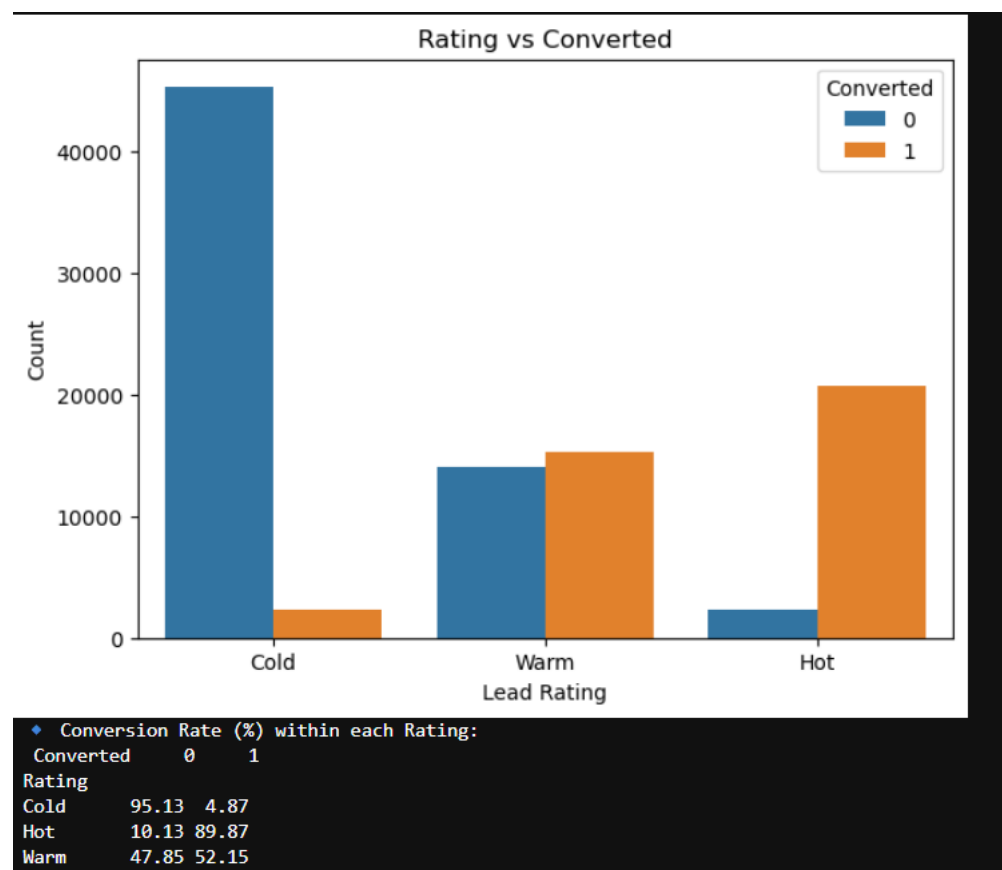
Here we'll see how the **lead rating** relates to **actual conversions**, and further examines how **product type** and **occupation** interact with both Rating and Conversion outcomes.

The objective is to:

Validate the effectiveness of the lead scoring model (Cold/Warm/Hot).

Identify which product categories and occupations contribute most to conversions.

Rating vs Converted



Visualization: Countplot of Rating (Cold, Warm, Hot) vs *Conversion Status* (Converted / Not Converted)

Observation:

There is a clear gradient in conversion success across ratings:

Cold Leads: Mostly Not Converted.

Warm Leads: Roughly balanced between Converted and Not Converted.

Hot Leads: Vast majority Converted.

Conversion Rate by Rating (%):

Rating	Converted (%)	Not Converted (%)
Cold	4.85	95.15
Warm	52.12	47.88
Hot	89.81	10.19

Insight:

Conversion probability increases sharply with lead rating.

Hot leads have the **highest conversion rate (≈90%)**, validating that the rating system is accurately capturing customer intent and readiness.

Warm leads have substantial potential (≈52%) and represent a valuable mid-tier segment.

Cold leads convert very rarely (~5%), indicating low immediate value.

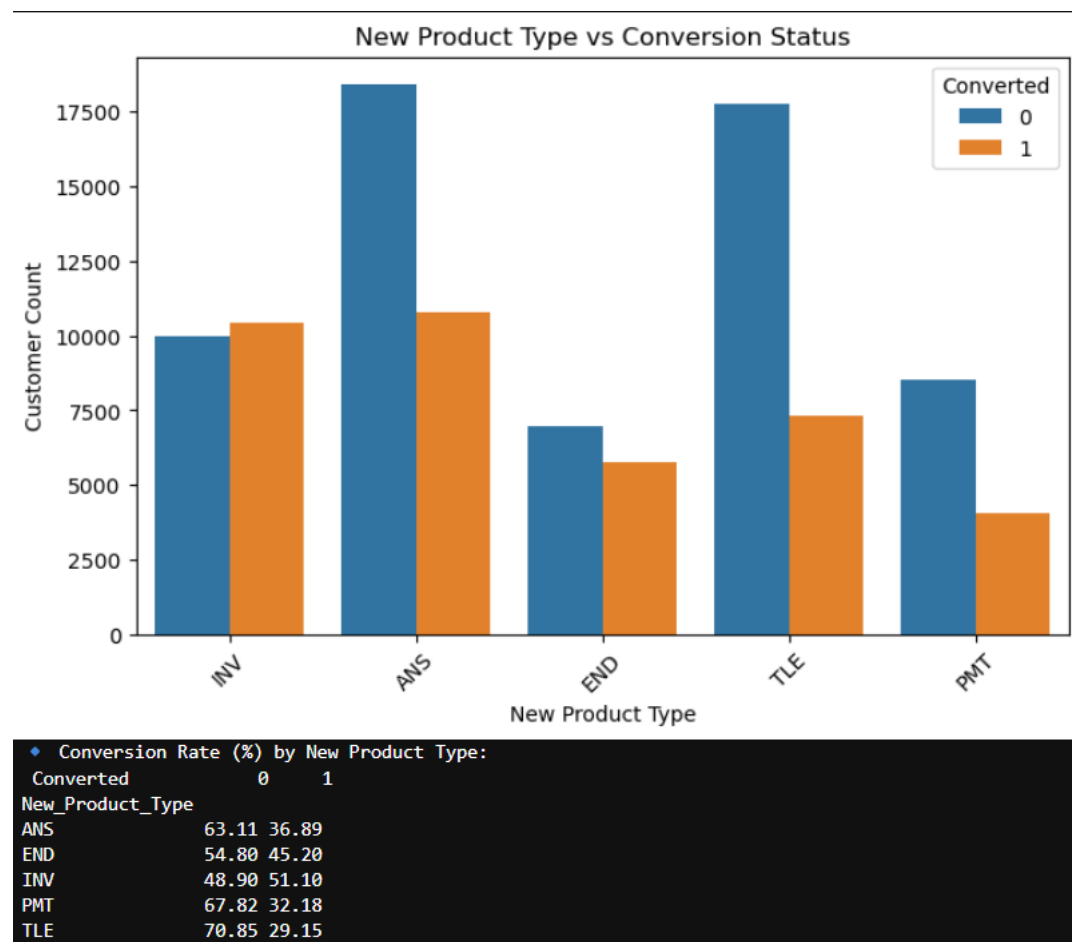
Business Implication:

Hot leads should be prioritized for **immediate sales action** (calls, personalized follow-ups).

Warm leads should be targeted through **email or remarketing campaigns** to maximize conversion potential.

Cold leads can be **nurtured** using long-term, low-cost engagement channels.

New Product Type vs Conversion



Visualization: Countplot comparing New Product Type with Conversion Status.

Observation:

Product types **INV (Investment)** and **END (Endowment)** dominate among converted customers.

ANS (Annuity) and other niche products have lower conversion shares.

Conversion Rate by Product Type (%):

New Product Type	Converted (%)	Not Converted (%)
INV	51.07	48.93
END	45.17	54.83
ANS	36.85	63.15
Others	<35	>65

Insight:

Investment-related products (**INV**) perform best, followed closely by Endowment (**END**).

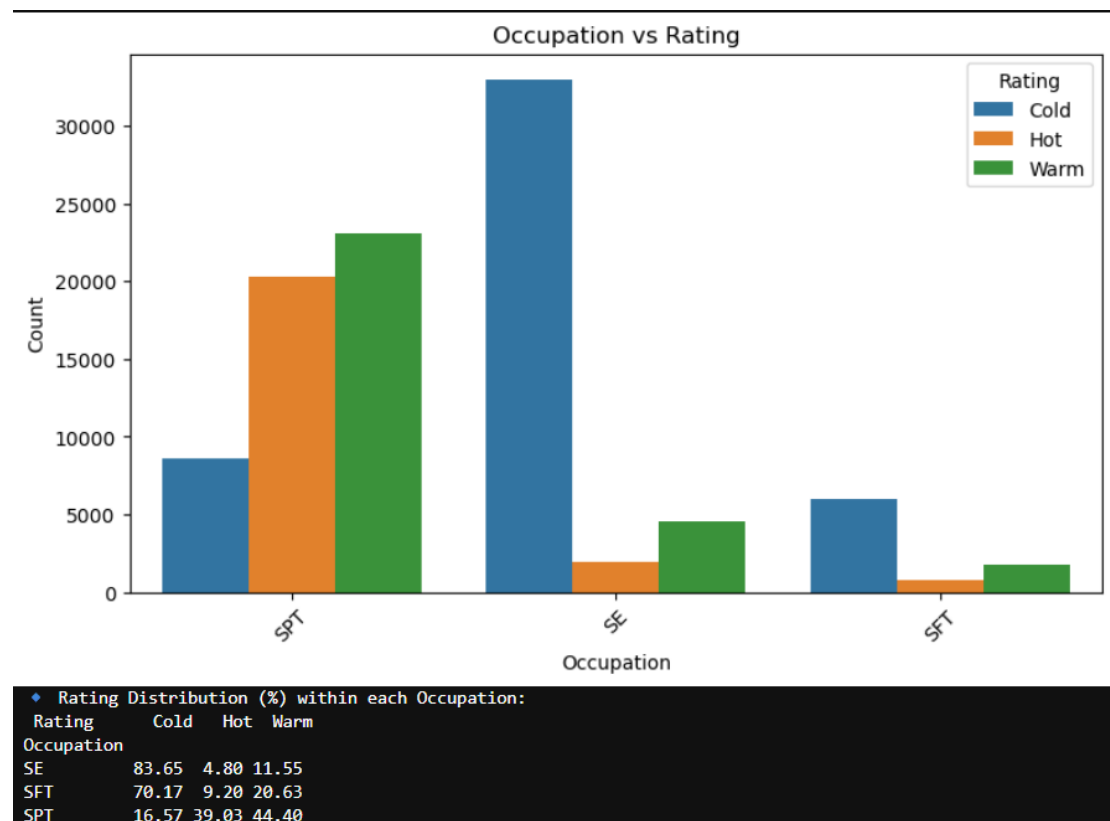
These categories represent **financial growth-oriented policies**, which appeal more to well-educated and higher-income customers.

Business Implication:

Focus on **Investment** and **Endowment** product lines in future marketing campaigns.

Consider rebranding or repositioning **Annuity** products to appeal to a younger audience, as they show lower conversion.

Occupation vs Rating



Visualization: Countplot of Occupation vs Rating, showing the rating distribution within each occupation.

Observation:

Occupations such as **SPT (Support Professionals)** and **SE (Self-Employed)** have the **highest proportion of Hot leads**, indicating stronger sales potential.

SFT (Service Field Technicians) dominate Cold leads, while **SLS (Sales)** and **ADM (Administrative)** roles tend to be Warm.

Rating Distribution by Occupation (%):

Occupation	Cold (%)	Warm (%)	Hot (%)
SFT	70.21	25.12	4.67
SE	16.33	37.87	45.80
SPT	12.04	28.07	59.89
ADM	25.55	47.50	26.95
SLS	24.17	54.18	21.65

Insight:

SPT and **SE** occupations are highly promising lead groups, with **Hot lead percentages exceeding 45–60%**.

SFT category primarily consists of Cold leads, showing low readiness to purchase.

Administrative and Sales categories show moderate Warm ratings, indicating mid-level potential.

Business Implication:

Prioritize **Support Professionals (SPT)** and **Self-Employed (SE)** customers for premium and investment-linked insurance offers.

Allocate lower marketing budgets to **Service Technicians (SFT)**, as they are less likely to convert.

Warm-rated Sales and Admin leads can be targeted with moderate-effort campaigns (email, phone, or online outreach).

Conclusion

In this project, we analyzed an insurance dataset to understand what factors influence customer conversions and to check how well the lead rating system (Cold, Warm, Hot) reflects actual outcomes.

We started by cleaning and preparing the data — handling missing values, standardizing formats, and ensuring the dataset was ready for analysis. Then, through a detailed exploratory analysis, we studied both numerical and categorical features to see how customer demographics, income levels, occupations, and product preferences affect their likelihood of purchasing a policy.

Our analysis showed that **younger, higher-income, and underinsured customers** are much more likely to convert, especially those interested in **investment and endowment-type products**. The **lead rating system proved to be very effective**, as conversion rates increased steadily from Cold to Warm to Hot leads.

Overall, the findings highlight that the company's existing rating model is reliable and can be used confidently for prioritizing leads. By focusing more on **Hot and Warm**

leads, targeting **professionals and self-employed customers**, and promoting **investment-oriented insurance plans**, the company can improve sales efficiency and customer engagement.

This analysis provides a clear data-backed direction for optimizing marketing strategies and achieving better conversion results.